

dynamically and therefore control the behaviour of the network as a whole. SDN is defined by two characteristics—the decoupling of the control and the data planes, and programmability on the control plane.

SDN is an emerging networking paradigm that provides a strong base to change the limitations of the current network infrastructure. First, it breaks the vertical integration by separating the network's control logic (the control plane) from the underlying routers and switches that forward the traffic (the data plane). Second, with the separation of the control and data planes, network switches become simple forwarding devices and the control logic is implemented in a logically centralised controller (or network operating system), simplifying policy enforcement and network reconfiguration and evolution.

## The SDN architecture

The architecture of SDN is highly dynamic, manageable, cost-effective and adaptable, and hence is well suited to high-bandwidth, dynamic applications. SDN architectures decouple network control and forwarding functions, enabling the former to become directly programmable, so all IT infrastructure gets abstracted from applications and network services.

## Characteristics of SDN architecture

- **Directly programmable:** All the network control can be programmed directly because it is decoupled from forwarding functions.
- **Agile:** Because of abstract control, network administrators can perform dynamic adjustments to the network-wide flow of traffic to make the desired changes.
- **Centralised management:** All the network management is centralised in SDN controllers and maintains a global view of the network, which appears to applications and policy engines as a single and logical switch.
- **Full configuration via programs:** SDN facilitates network administrators in configuring, managing, securing and optimising network resources very efficiently through dynamic and automated SDN programs.
- **Based on open standards and freedom from vendors:** SDN facilitates simple network design and all operations, because the operation instructions are provided by SDN controllers rather than multiple, vendor-specific devices and protocols.

## Components of SDN architecture

- **SDN applications:** These are the programs that explicitly, directly and programmatically communicate their network requirements and desired network behaviour to the SDN controller through the northbound interface (NBI).
- **SDN controller:** This is a logical entity that receives instructions or requirements from the SDN application layer and relays them to the networking components. It consists of one or more NBI agents, the SDN control logic

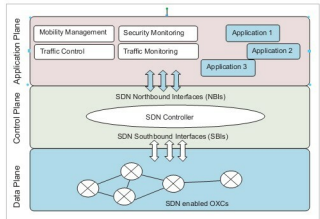


Figure 1: SDN architecture

and the 'control to data-plane interface' (CDPI) driver.

- **SDN northbound interfaces (NBI):** These are the interfaces between SDN applications and SDN controllers. They provide abstract network views and enable direct expression of network behaviour and requirements.
- **SDN southbound interfaces (SBI):** This is an OpenFlow protocol specification. Its main function is to enable communication between the SDN controller and the network nodes (physical and virtual switches and routers) to facilitate the router in discovering network topology, in defining network flows and relaying requests to the northbound APIs.
- **SDN data path:** This is a logical network device that exposes visibility and uncontested control over its advertised forwarding and data processing capabilities. It consists of the CDPI agent, a set of one or more traffic forwarding engines, and zero or more traffic processing functions.
- **SDN control to data-plane interface (CDPI):** This is the interface between the SDN controller and SDN data path to provide programming control of all forwarding operations, advertisements, statistical reporting and the notification of events.

## The planes in SDN

SDN spans five different planes:

- Forwarding plane
- Operational plane
- Control plane
- Management plane
- Application plane

In addition to the above, there are two abstraction layers—the device and resource abstraction layer (DAL) and the network service abstraction layer (NSAL). There are also two interfaces—one to facilitate communication between the control plane, the forwarding plane and the management plane; and the other between the management plane and the operation plane.